

MAIN-BELT ASTEROID

1 Ceres

AVERAGE DISTANCE TO SUN 257 million miles (414 million km)

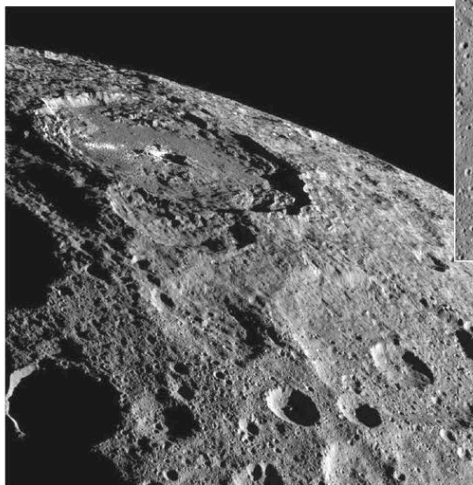
ORBITAL PERIOD 4.60 years

ROTATION PERIOD 9.08 hours

DIAMETER 590 miles (950 km)

DATE OF DISCOVERY January 1, 1801

Ceres was discovered by accident in 1801 by Giuseppe Piazzi, the director of the Palermo Observatory in Italy, while he was compiling a catalog of fixed stars (see panel, right). One of the “stars” had moved during the night, and this turned out to be the first



LANDSCAPES

Ceres landscape varies across its surface. A 57-mile (92-km) wide crater dominates the view on the left. Bright spots on its floor are probably salts left after a briny liquid oozed up from below. The 22-mile (35-km) wide view on the right contains a series of extensional fractures.

known asteroid, Ceres. Some 100 years before, Johannes Kepler (see p.68) had suspected that there was a “missing” planet in the gap between the orbits of Mars and Jupiter (see pp.170–171). By 1800, some of Europe’s leading astronomers had started to look for objects in this gap, and Piazzi made the first discovery. About a year later, a German doctor and astronomer, Heinrich Olbers, was observing the path of Ceres in an attempt to produce a more accurate estimate of its orbital parameters when he discovered a second asteroid. This

was given the number 2 and named Pallas. The orbits of Ceres and Pallas were found to cross, and Olbers concluded that they were fragments of a planet that had broken up. As the century progressed, more and more asteroids were discovered that were smaller and fainter than the first two. Using spectroscopes to analyze the light reflected from their surfaces, astronomers found that asteroids had different colors due to different compositions, and this led to the establishment of a classification system (see p.170). Ceres was classified as a carbonaceous, or C-type, asteroid.

In 2006, the new category of dwarf planets was introduced to describe objects that are rounded in shape but have not swept their orbits clear of other bodies (see p.209). Ceres was placed in this category. However, it remains the

largest member of the asteroid belt, so it can be said to have a dual identity as both a dwarf planet and an asteroid. NASA’s Dawn probe collected data from orbit about Ceres between April 2015 and November 2018, after previously studying Vesta (see opposite).

EXPLORING SPACE

PIAZZI’S TELESCOPE

Known as the Palermo Circle, this telescope was made between 1787 and 1789 by Jesse Ramsden of London, England, the greatest European instrument maker of the 18th century. Its lens has an aperture of 3 in (75 mm). The circular altitude scale and the horizontal azimuth scale are both read using microscopes. In its day, it was the most southerly European telescope, being on top of the Royal Palace in Palermo, Sicily. While measuring star positions with this telescope, Piazzi discovered the first asteroid, Ceres.



NEAR-EARTH ASTEROID

25143 Itokawa

AVERAGE DISTANCE TO SUN 123 million miles (198 million km)

ORBITAL PERIOD 1.52 years

ROTATION PERIOD 12.1 hours

LENGTH 0.34 miles (0.54 km)

DATE OF DISCOVERY September 26, 1998

Itokawa is a small, irregularly shaped asteroid of the type known as a “rubble pile,” meaning that it is not a solid, coherent body. The Japan Aerospace Exploration Agency (JAXA) chose it as the target for a sample-collection mission called

Hayabusa, which the agency launched in May 2003 (see panel, below). At the time of launch, the asteroid bore only the reference number 1998SF36, but it was named in honor of Hideo Itokawa (1912–1999), known as the father of Japanese rocketry, while the probe was on its way.

The Hayabusa probe reached Itokawa in September 2005 and spent 2 months surveying the asteroid before attempting a landing. The asteroid’s picklelike shape, 1,110 ft (540 m) long and 690 ft (210 m) across at its narrowest, gives it the appearance of two separate masses stuck together. Astronomers think that Itokawa was once much bigger, possibly up to 12 miles (20 km)

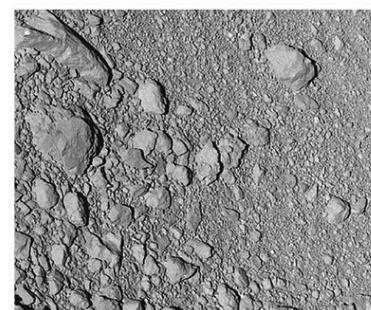


across. Large impacts broke the asteroid into smaller pieces, which then gently reassembled to form the low-density rubble-pile structure that we see today.

Rocks that are up to 165 ft (50 m) wide are dotted over Itokawa’s surface, while the narrow neck near the middle is smoother and covered with dust. In November 2005, the Hayabusa space probe gently touched down on the smooth part. It collected numerous microscopic particles of the asteroid’s dust, which it brought back to Earth. The sample return capsule blazed through the Earth’s atmosphere to land at Woomera, Australia, in June 2010. The capsule was retrieved and opened under sterile conditions in a laboratory in Japan. The asteroid’s surface rocks proved to be rich in the mineral olivine, similar to common types of chondrite meteorites.

ELONGATED ASTEROID

This image of Itokawa was taken by the Hayabusa probe, which landed in the smooth area near the middle to collect dust samples. The asteroid lacks obvious impact craters, in contrast to most others visited by spacecraft.



ROUGH SURFACE

Large rocks can be seen strewn over Itokawa’s surface in this close-up. These rocks are probably fragments of an earlier breakup of the asteroid that have since collected together again.

EXPLORING SPACE

THE HAYABUSA MISSION

Hayabusa was a Japanese probe sent to rendezvous with and bring back samples from the near-Earth asteroid Itokawa. Hayabusa is the Japanese name for the peregrine falcon. It was intended to swoop down on the asteroid like a hawk, to take samples, and then return them to Earth. It spent 30 minutes on the asteroid’s surface before taking off again with dust samples.

ASTEROID APPROACH

An artist’s impression shows the Japanese space probe Hayabusa approaching asteroid Itokawa in November 2005.

